PAYMENT TRANSACTION MANAGEMENT SYSTEM

1. INTRODUCTION

The Payment Transaction Management System is developed using MySQL to manage customer payment transactions. It stores payment details such as payment date, payment mode, transaction amount and payment status. It helps to track successful, failed and pending payments and generate revenue reports.

2. OBJECTIVES

- To maintain customer payment records.
- To connect payments with existing orders.
- To manage different payment methods.
- To update payment transaction status.
- To identify successful, failed and pending payments.
- To generate revenue and payment analysis reports.

3. PAYMENT TABLE DESIGN

Table Name: Payment

Column Name| Data Type| Constraint
Payment_ID| INT| Primary Key
Order_ID| INT| Foreign Key, Unique
Payment_Date| DATE| Not Null
Payment_Mode| VARCHAR(30)| Not Null
Payment_Status| VARCHAR(20)| Not Null
Transaction_Amount| DECIMAL(10,2)| Not Null

Payment Modes:

- UPI
- Credit Card
- Debit Card
- Net Banking
- Cash on Delivery

Payment Status:

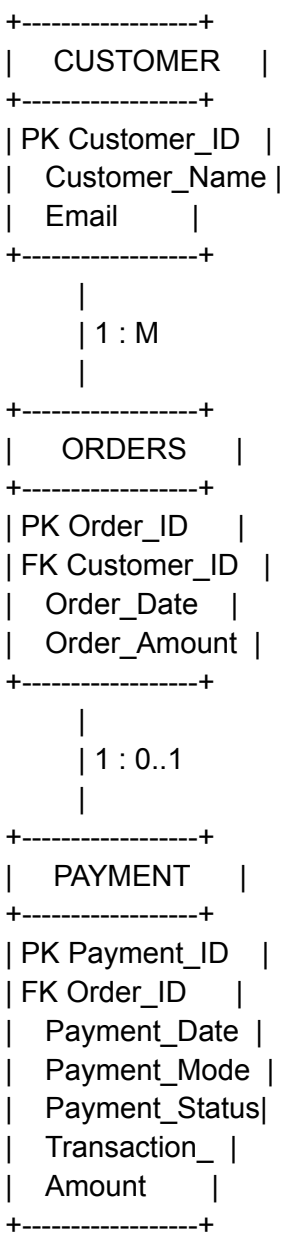
- Success
- Failed
- Pending

4. UPDATED ER DIAGRAM

The Payment table is connected to the Orders table through Order_ID.

Relationship:

- One Customer can have many Orders.
- Each Order can have at most one Payment.
- Each Payment belongs to one Order.



5. SAMPLE CUSTOMER DATA

Customer_ID| Customer_Name| Email
1| Arun| arun@gmail.com
2| Priya| priya@gmail.com
3| Kumar| kumar@gmail.com
4| Divya| divya@gmail.com
5| Ravi| ravi@gmail.com

6. SAMPLE ORDER DATA

Order_ID| Customer_ID| Order_Date| Order_Amount
101| 1| 2026-10-01| 2500
102| 2| 2026-10-02| 3500
103| 3| 2026-10-03| 1500
104| 4| 2026-10-04| 4000
105| 5| 2026-10-05| 2000

7. SAMPLE PAYMENT DATA

Payment_ID| Order_ID| Payment_Mode| Payment_Status| Amount
1| 101| UPI| Success| 2500
2| 102| Credit Card| Failed| 3500
3| 103| Debit Card| Success| 1500
4| 104| Net Banking| Pending| 4000
5| 105| Cash on Delivery| Success| 2000

8. PAYMENT TRANSACTION OPERATIONS

The following operations are performed:

1. Insert new payment transactions.
2. Display successful payments.
3. Find failed and pending payments.
4. Update payment status after retry.
5. Retrieve payment details for a specific order.
6. Count payments based on transaction status.
7. Display complete payment details with customer and order information.

9. PAYMENT TRANSACTION AND REVENUE ANALYSIS REPORTS

The system generates the following reports:

1. Transactions by Payment Mode

Displays the number of transactions made using each payment method.

2. Most Preferred Payment Method

Identifies the payment method used most frequently.

3. Total Successful Payment Revenue

Calculates the total amount received from successful transactions.

4. Revenue by Payment Method

Displays the successful revenue collected through each payment method.

5. Average Transaction Amount

Calculates the average amount of all payment transactions.

6. Customer Payment History

Displays the payment history of customers along with their order details.

7. Payment Status Report

Displays the total number of successful, failed and pending transactions.

8. Failed and Pending Orders

Identifies orders that have failed or pending payments.

10. QUERY OUTPUT SCREENSHOTS

Attach screenshots captured from MySQL Workbench for the following queries:

1. Payment table creation successfully executed.
2. Sample payment records inserted.
3. Successful payment details.
4. Failed and pending payment details.
5. Updated payment status.
6. Payment transactions by mode.
7. Total successful payment revenue.
8. Revenue by payment method.
9. Customer payment history.

10. Complete payment information with customer and order details.

11. ADVANTAGES

- Easy management of payment records.
- Reduces manual work.
- Maintains accurate transaction details.
- Helps track payment status.
- Generates useful revenue reports.
- Provides better order and payment management.

12. CONCLUSION

The Payment Transaction Management System successfully manages payment transactions using MySQL. It maintains customer, order and payment details with suitable constraints and relationships. It also provides payment status tracking and revenue analysis reports. Thus, the system improves the accuracy and efficiency of payment management.